

Mental Readiness and Fatigue: Extended Perspectives on Cognitive Functioning and Flow Experiences at Work

Lukas Schick
Institute for Information Systems
Karlsruhe Institute of Technology
(KIT)
Karlsruhe, Germany
lukas.schick@kit.edu

Michael Thomas Knierim
Institute for Information Systems
Karlsruhe Institute of Technology
(KIT)
Karlsruhe, Germany
michael.knierim@kit.edu

Benjamin Gaube
Institute for Information Systems
Karlsruhe Institute of Technology
(KIT)
Karlsruhe, Germany
benjamin.gaube@kit.edu

Mario Nadj
Rhine-Ruhr Institute of Information
Systems
University of Duisburg-Essen
Essen, Germany
mario.nadj@ris.uni-due.de

Alexander Mädche
human-centered systems lab (h-lab)
Karlsruhe Institute of Technology
(KIT)
Karlsruhe, Germany
alexander.maedche@kit.edu

Abstract

Mental fatigue and mental readiness reflect key aspects of cognitive functioning in everyday knowledge work. Fatigue indicates diminished concentration, while readiness captures a positive state of preparedness influenced by mood, motivation, and alertness. Understanding their interplay may be crucial for enhancing well-being in digital work environments. We conducted a two-week-long experience sampling study in a home-office setting, where eight participants alternated between controlled tasks and natural knowledge work over ten sessions. Results show that mental fatigue and readiness are inversely related but distinct, each contributing uniquely to mental workload and flow experiences. Readiness predicts higher workload, reflecting increased engagement when individuals feel motivated and alert. Flow strongly correlates with post-task readiness, suggesting it maintains positive motivational aspects, even amid increased fatigue. These findings underscore the nuanced roles of cognitive states in shaping well-being and highlight opportunities for adaptive systems to foster positive mental states in real-world work contexts.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*.

Keywords

Mental Fatigue, Mental Readiness, Mental Workload, Flow Experience, Knowledge Work

ACM Reference Format:

Lukas Schick, Michael Thomas Knierim, Benjamin Gaube, Mario Nadj, and Alexander Mädche. 2025. Mental Readiness and Fatigue: Extended Perspectives on Cognitive Functioning and Flow Experiences at Work. In

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719836>

Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719836>

1 Introduction

Knowledge work today is inseparable from digital technologies, which have revolutionized productivity and connectivity. However, this transformation also demands constant attention, availability, and faster working speeds, leading to increased cognitive demands, stress, and diminished mental well-being [44]. For instance, the rise of video conferencing platforms like Zoom has been shown to exacerbate mental demands, creating unprecedented levels of mental fatigue during the workday [35], which can result in occupational burnout [13]. To address these challenges, HCI researchers are developing intelligent technologies that leverage subjective measures (e.g., self-assessment questionnaires) and objective biosignals to monitor mental states, which can be used to give feedback or to adapt systems [1, 38].

While mental fatigue highlights the negative side of cognitive functioning, it represents only one side of the well-being coin. Recently, the concept of mental readiness has gained attention as a complementary perspective. Mental readiness refers to the subjective feeling of being prepared to tackle a specific task and is influenced by external factors such as sleep quality, as well as internal factors like mood and motivation [26, 28, 30, 31]. Like mental fatigue, mental readiness has been linked to task performance and mental well-being. Yet, unlike mental fatigue, mental readiness emphasizes the potential for positive cognitive states.

However, up to now, no research has been conducted to systematically assess whether mental readiness and fatigue represent synonymous constructs or complementary perspectives, especially in the context of everyday work life. Importantly, it has not been studied yet, if the two constructs allow explaining the development of typical experiences like mental workload during work.

In addition, as the two constructs capture different underlying dimensions like motivation and concentration, it would appear plausible that both constructs may be related to well-being conducive experiences like flow.

Flow can be experienced in knowledge work scenarios [10, 32, 36, 49], is considered to require both concentration and motivation to enter it [11], and has been found to occur less when recovery from work has been poor [12]. For this reason, it appears as a prime candidate to assess whether mental readiness, beyond the absence of concentration difficulties, relates to the emergence of flow experiences at work. Therefore, to advance the understanding of how mental fatigue and readiness relate to each other and how they can inform the occurrence of typical, well-being related experiences at work, we set out to explore the following research questions (RQ):

- (1) Are mental readiness and fatigue complementary and separate constructs?
- (2) How do mental readiness and fatigue develop in everyday knowledge work settings?
- (3) How do mental readiness and fatigue relate to workload and flow experiences?

To answer these RQ, a semi-controlled experience sampling (ESM) field study was conducted, mimicking a home-office work setting. Over the course of ten sessions in two weeks, eight participants worked on both controlled mental arithmetic tasks (as a reference point), and on their own natural knowledge work projects, providing repeated self-reports of their pre-, mid-, and post-task experiences.

Our results show that there exists a strong and significant inverse relationship between mental readiness and fatigue, and that both constructs exhibit the expected changes across work sessions (readiness decreases and fatigue increases after work). However, importantly, factor analysis results show that they should nonetheless be considered as separate constructs and both constructs show diverging relationships with mental workload and flow. Mental readiness (but not mental fatigue) shows a significant positive predictive relationship with mental workload, indicating that with higher readiness, more workload is experienced in the ensuing task. Furthermore, flow shows a robust positive correlation with post-task mental readiness. At the same time, we also find a weaker indication that flow might be positively related with post-task mental fatigue. These findings align with existing literature suggesting that flow can sustain motivation and prevent burnout [3], despite potentially increasing mental fatigue due to high cognitive demands [21]. Our results hint that flow might maintain positive motivational aspects of readiness while not entirely mitigating fatigue, potentially through a buffering effect that offsets the negative impact of fatigue with positive connotations.

Our results, therefore, contribute to the HCI-literature by addressing an important gap in the understanding of the everyday negative and positive dimensions of cognitive functioning. These findings can contribute to a better understanding of typical workload experiences during a day, providing possible extensions of previous HCI-work on the daily mental workload cycle [29]. Our results also highlight new opportunities for investigating everyday experiences through combining mental readiness and fatigue assessments. We close this contribution with a discussion of how future technologies might try to capitalize on detecting changes in readiness and fatigue, for example through the use of wearable biosensors.

2 Related Work

Mental readiness is defined as an optimal mental state for performing a given task and originates in sports psychology [25, 31]. Motivation, focus and attention on the task, and mood are considered as main aspects of mental readiness, while the use of mental preparation techniques is further highlighted [30, 31]. Mental readiness was subsequently applied to members of other professions like dentists [28] and surgeons, where a significant positive correlation between mental readiness and performance was found [27]. Further studies show links between high mental readiness levels and reduced stress as well as increased mental well-being at work [12, 25, 26]. Little is known on the impact of mental readiness on mental well-being and performance in digital knowledge work. Debus et al. conducted a study among software engineers to investigate the influence of being “recovered” from job stress on work-related flow. While not the same concept, it shares many characteristics with mental readiness. The study found a positive relationship between the degree of recovery and flow states at work during the day [12]. A study by Ng et al. focused on the connection between sleep quality and mental readiness, finding a link between inadequate sleep and decreased mental readiness. However, no potential links between mental readiness and its impact on knowledge work were investigated [30]. So, while related concepts have been studied, the impact of mental readiness on performance and mental well-being in knowledge work, and its relationship to the established concepts of mental fatigue, mental workload and flow, has yet to be investigated.

Contrary to mental readiness, the concepts of mental fatigue, mental workload and flow are well established in the knowledge work context [1, 2, 36, 48, 49]. Conceptually, mental fatigue seems most closely linked to mental readiness. While mental readiness is characterized as a mental state of high focus, positive mindset and motivation, mental fatigue is described as one of distractedness, lack of energy, and disinterest [8, 19], suggesting a strong, inverse relationship between the two. The negative impact of mental fatigue on performance and mental well-being in knowledge work is well documented [34, 42]. The main cause of mental fatigue are prolonged periods of high mental workload [9, 15]. Mental workload is often described as the cognitive strain the execution of a certain task puts on a person. This depends on the demands of the task, the skill level of the person working on the task and the available cognitive resources [45]. In the knowledge work domain, mental workload is often studied to measure the impact of technologies and human-machine-interaction on the mental state and performance of workers [16, 41, 43].

The final concept whose relationship to mental readiness is studied in this paper is flow [11]. Flow is described as a mental state in which the level of challenge of a task is matched by the person’s capabilities. This leads to an increased enjoyment of working on the task, and is associated with increased feelings of alertness, happiness, and satisfaction [11, 17]. There is even evidence that flow can protect workers from burnout [3]. As being in a state of flow is generally regarded as desirable, a lot of research into flow in the knowledge work context examines how flow can be fostered [5–7, 49]. A review of flow literature conducted by Peifer et al. hints at a possible connection between flow and mental readiness. Their

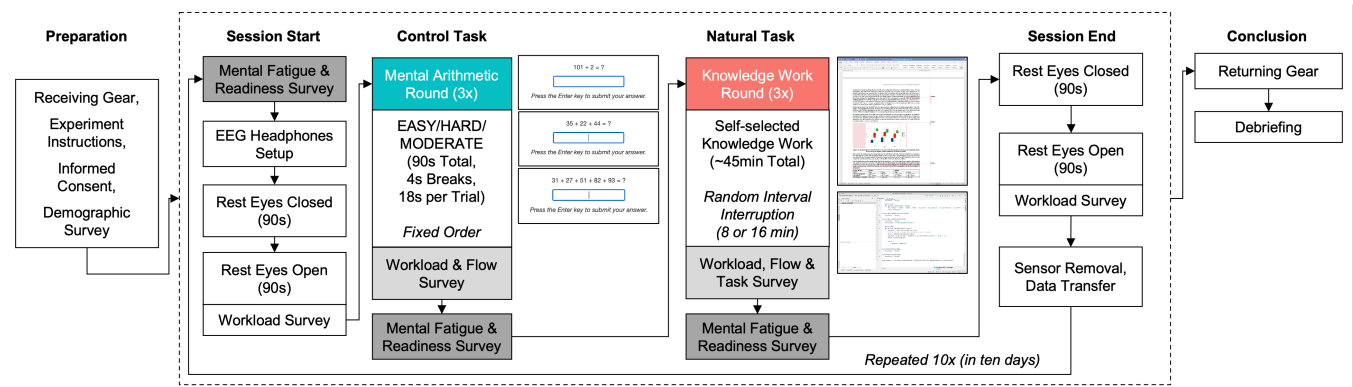


Figure 1: Study procedure overview. The key questionnaire moments are shown in gray. The two main tasks in blue and red.

results show that, among other factors, focus, motivation and positive mood all positively influence the likelihood of flow [33], which are all inherent factors of mental readiness as well.

3 Method

3.1 Procedure, Tasks, and Measures

We conducted a controlled ESM field study combining controlled tasks as reference points with open, naturalistic knowledge work tasks (e.g., programming or scientific writing). This design allowed for comparisons between homogenized repeated and heterogeneous natural experiences, balancing internal and ecological validity at the within-subject level. The complete study design is visualized in Figure 1.

In the controlled mental arithmetic task, participants added 2, 4, and 6 summands in three task conditions (easy, medium, and hard), respectively. The design is closely aligned with similar studies [20, 22] and chosen as an established approach for eliciting mental workload to varying degrees. This variation helps verify that mental fatigue changes consistently across conditions, making the pre- and post-task fatigue measures more robust.

In contrast, the naturalistic tasks focused on the open choice of knowledge work tasks similar to related studies [37, 46] and varied between participants, sessions, and even within sessions. Thus, to enhance reliability, data were collected over ten 45-minute home-office sessions across ten days for each participant. The response rate for the ESM prompts was 100% across the ten days of data collection.

During each session, the perceptions of mental readiness and mental fatigue were sampled before and after each task (i.e. controlled or natural work task). Mental fatigue was measured with the 4-item, 7P-Likert mental fatigue subscale of the multidimensional fatigue inventory - MFI by Smets et al. [39]. Mental readiness was measured with the 3-item, 100P-rating scale by Ng et al. [30]. The perceptions of mental workload and flow were sampled after each mental arithmetic condition and by repeatedly interrupting the natural work with surveys (randomized survey interruptions occurred after 8 or 16 minutes so that participants could not fully anticipate the next interruption), with three interruptions for each work session. Including multiple interruptions helps ensure a more

reliable and representative measure of mental workload and flow by reducing the impact of momentary fluctuations or outliers. A single questionnaire at the end might capture an unrepresentative snapshot influenced by factors like recency effects or task-ending impressions, whereas averaging multiple measurements provides a more stable and accurate estimate of the overall experience. Mental workload was measured with the NASA TLX 6-item, 21P-rating scale [18] and flow with the 10-item, 7P-Likert flow short scale (FKS) by Engeser et al. [14]. All questionnaires were recorded using their recommended scaling and answer formats to ensure their internal validity. All constructs showed good internal consistency through their Cronbach's Alpha values above 0.7 (mental readiness = 0.78, mental fatigue = 0.89, mental workload = 0.84, flow = 0.92).

Lastly, each work session was preceded and concluded by resting phases. These were added to provide a moment of rest to mentally prepare for the experiment and for the calibration of attached physiological sensors. Specifically, participants put on a pair of EEG headphones in every session to measure neural oscillations related to mental workload. In this article, we do not report on these data to maintain focus on establishing an understanding of the mental readiness and fatigue experiences.

3.2 Setup & Sampling

The study was approved by the first author's institutional review board. The data were collected from a convenience sample of 8 participants (2 female, 6 male, age range 20-30, mean age 26.1 years) who were recruited from the academic staff and students in the authors' institution and participated voluntarily without remuneration.

Participants were initially informed about the general structure of the experiment, including the hardware, software, and recording procedure. For the work sessions, they agreed to adhere to the following restrictions: the work during a recording session should be considered knowledge work, last for 45 minutes, and be conducted in isolation, without interruption, and without speaking or eating. At the start of the study, participants received a parcel with the recording system (the above-mentioned EEG headphones) and the study consent form to be signed and uploaded at the beginning of the study. From this point on, participants recorded the remainder of the experiment by themselves.

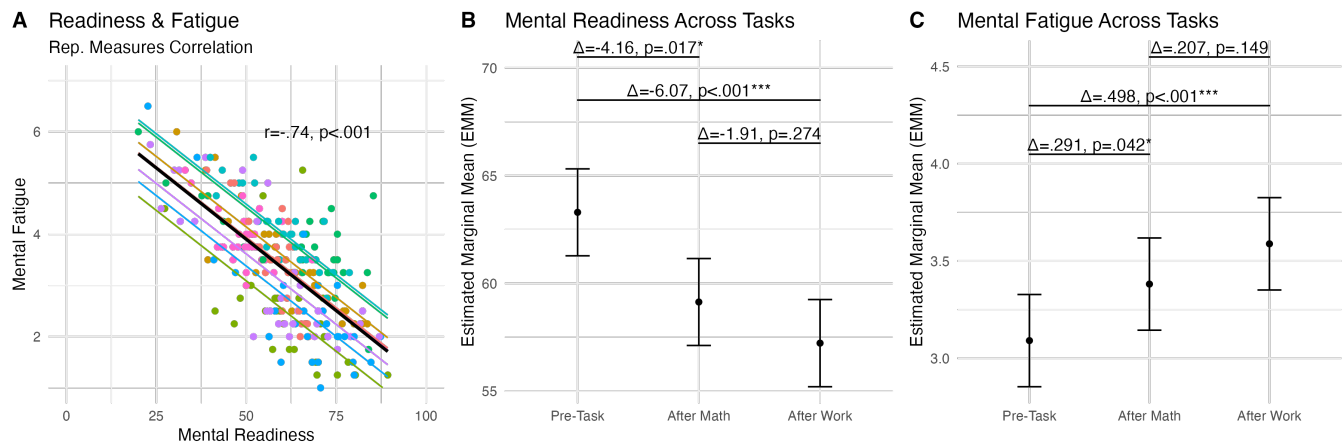


Figure 2: A: Rep. measures correlation of mental readiness and fatigue. Colors represent participants. B&C: Changes in mental readiness and fatigue across tasks.

4 Results

4.1 Relationship between Mental Readiness and Mental Fatigue

We analyzed the convergent and divergent validity of mental readiness and fatigue to better understand whether the constructs should be considered separate or whether they represent highly similar phenomena.

Initially, we assessed the correlations of the two constructs and whether they show the expected changes over the course of an experiment session. All reported correlations were computed as repeated measures correlations [4] to account for the nested data structure from repeated experiment sessions with the same subjects. For the same reason, mixed effects regression models with participants as random intercepts were used when constructing linear models.

A significant and strong negative correlation was found between mental readiness and fatigue ($r_{rmc} = -.74, p < .001$ - see Figure 2A), indicating a close, inverse relationship between both constructs.

Furthermore, a significant decrease in mental readiness and increase in mental fatigue from the start of the session is observed (see the estimated marginal means in Figure 2B&C). Thus, mental readiness and fatigue show a strong degree of convergent validity. To assess their divergent validity, a factor analysis was conducted.

Parallel analysis to determine the number of factors confirmed that two factors were best suited to describe mental readiness and fatigue.

An exploratory factor analysis with two factors yielded high factor loadings (all > 0.6) for the mental fatigue subscales with one factor, while all three subscales of mental readiness exhibited a high factor loading (all > 0.6) with the other factor. All other factor loadings between subscales and factors were considerably smaller, clearly pointing towards mental readiness and fatigue being two distinct concepts. The confirmatory factor analysis verified this conclusion, yielding a higher Comparative Fit Index (0.911 vs 0.862)

and Tucker-Lewis Index (0.857 vs 0.793), and a lower root-mean-square error of approximation (0.179 vs 0.215) for the two-factor model (see also Figure 3 for the items and factor loadings).

Thus, altogether, these results suggest that it is better to consider mental readiness and fatigue as two separate constructs.

4.2 Development of Mental Readiness and Fatigue during Knowledge Work

Significant differences in the estimated marginal mean between the start and the end of the experiment session have been found for both mental readiness and mental fatigue (both $p < .001$).

As can be seen in Figure 2B&C, the estimated marginal mean for mental readiness is highest at the start of the experiment session. It subsequently decreases and reaches its lowest value after the participants have completed both tasks. Mental fatigue shows an opposing trend. With the lowest estimated marginal mean reported at the beginning of the experiment, it increases and reaches its highest value at the end of the experiment session. Note that the mental arithmetic task appears to have a stronger effect than the natural task, despite its shorter duration. However, both mental readiness and mental fatigue show their lowest and highest levels, respectively, at the end of the session.

These trends follow established patterns in the literature, as with periods of cognitively demanding work, aspects of mental readiness such as focus and attention are expected to decrease, while mental fatigue is expected to increase over time [8, 12].

4.3 Relationship of Mental Readiness to Mental Workload and Flow

Next, to understand if and how mental readiness and fatigue related differently to well-studied (mental workload) and less often studied (flow) phenomena, a series of correlation analyses were conducted - investigating both (1) how readiness and fatigue at the start of the task predict ensuing experience, and (2) how task experiences impact mental readiness and fatigue after the task.

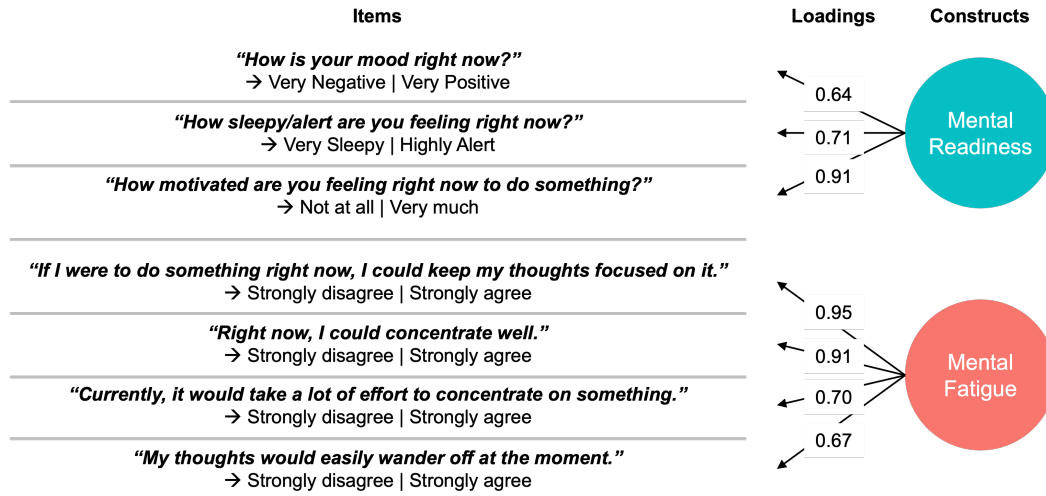


Figure 3: CFA-based 2-factor map of mental readiness and fatigue, showing the items and their respective factor loadings.

These analyses were conducted for the mental arithmetic task and the work task separately, especially as we expect higher comparability of the effects in the mental arithmetic task and want to use this as a reference to the natural work observations.

For the former set of analyses (predicting workload and flow from readiness and fatigue before the task), the variables were participant-centered by subtracting each participant's mean response. For the latter set of analyses (predicting final readiness and fatigue from workload and flow), the readiness and fatigue values were session-centered instead. These centering decisions focus on isolating within-subject changes per session. Thus, the results ought to be interpreted as changes across sessions, not as changes across participants.

A summary of the significant correlation results is displayed in Figure 4.

4.3.1 Mental Workload Relationships. In the mental arithmetic task, the pre-task mental readiness level showed a significant, positive correlation to the average mental workload during the sessions ($r_{\text{rmc}}=.30$, $p=.017^*$). This effect is similar for the natural work task ($r_{\text{rmc}}=.20$, $p=.116$), but weaker and no longer significant, which could be due to the higher experience heterogeneity from the natural work.

Pre-task mental fatigue showed no significant relationship with ensuing mental workload levels in either the mental arithmetic task ($r_{\text{rmc}}=-.01$, $p=.957$) or the natural work tasks ($r_{\text{rmc}}=.11$, $p=.385$). Therefore, readiness, but not fatigue, showed predictive power for following mental workload.

For the post-task mental readiness assessment, no significant relationships between the average mental workload were found in either the mental arithmetic ($r_{\text{rmc}}=.23$, $p=.052$) or the natural work task ($r_{\text{rmc}}=-.05$, $p=.677$). Similarly, no significant relationships were found between post-task mental fatigue and mental workload in either task (Mental Arithmetic: $r_{\text{rmc}}=-.12$, $p=.312$, Work: $r_{\text{rmc}}=-.06$, $p=.594$).

This means that when a participant experienced more mental workload in a session, it did not result in relatively more or less fatigue or readiness for himself/herself.

4.3.2 Flow. No significant relationships were found between pre-task mental readiness and flow experiences (Mental Arithmetic: $r_{\text{rmc}}=.21$, $p=.088$, Work: $r_{\text{rmc}}=-.04$, $p=.765$) and also no significant relationships between pre-task mental fatigue and ensuing flow experiences (Mental Arithmetic: $r_{\text{rmc}}=.20$, $p=.116$, Work: $r_{\text{rmc}}=-.02$, $p=.874$).

However, average flow during both tasks was significantly positively correlated with mental readiness after the task (Mental Arithmetic: $r_{\text{rmc}}=.30$, $p=.009^{**}$, Work: $r_{\text{rmc}}=.30$, $p=.010^*$), indicating that when a participant experienced more intense flow, their mental readiness after the session was relatively higher than in other sessions.

Furthermore, a significant positive correlation was also found between average flow and the mental fatigue level after the mental arithmetic task ($r_{\text{rmc}}=.23$, $p=.048^*$), indicating that when more intense flow was experienced in a session, also relatively more fatigue resulted. Once more, this effect is similar for the natural work task ($r_{\text{rmc}}=.17$, $p=.166$), but weaker and no longer significant, which could be due to the higher experience heterogeneity from the natural work.

5 Discussion

The factor analysis carried out in this study shows that, while a clear inverse linear relationship exists between mental readiness and fatigue, they are separate concepts. The relationships of mental readiness and fatigue to workload and flow further highlight that, in addition to being separate concepts, they are also complementary concepts. For the mental arithmetic task, mental readiness pre-task has a significant positive effect on mental workload during the task, indicating that when a participant feels more ready to work on the task, then they may be prepared to expend more mental

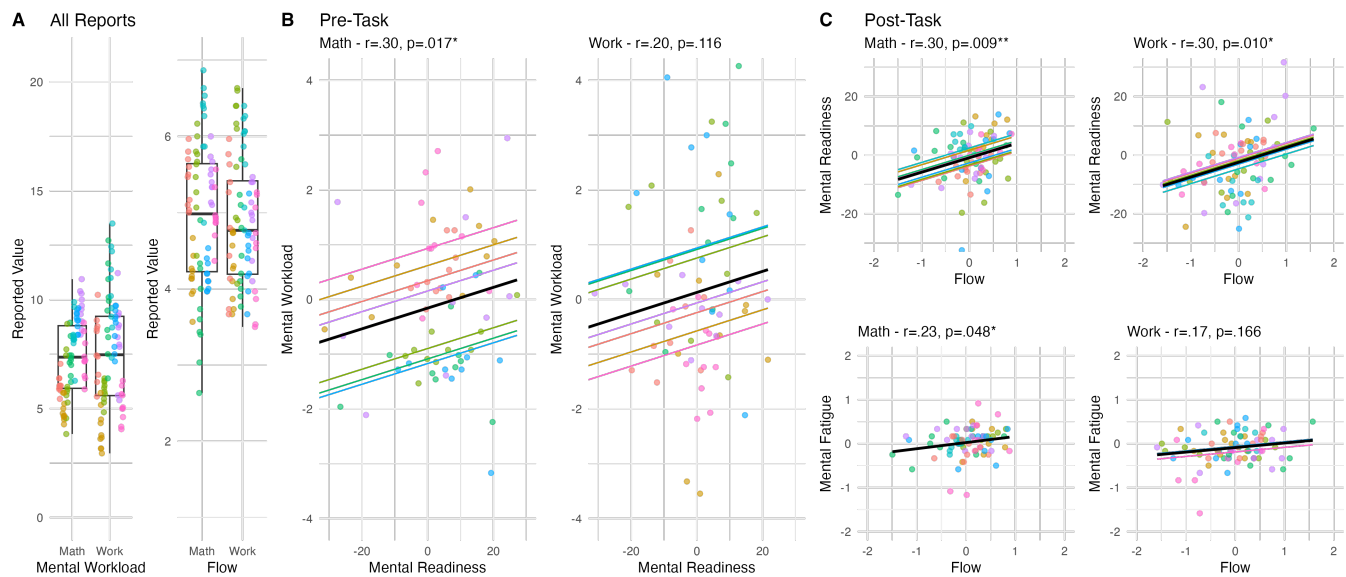


Figure 4: A: Reported mental workload and flow. B: Pre-Task correlations with mental readiness and fatigue. C: Post-Task correlations. Colors represent participants.

effort on this task. A possible explanation might be the high motivation associated with mental readiness, which may lead participants to spend more cognitive resources on a task than they otherwise would. No correlation was found between mental workload during the tasks and mental readiness or fatigue post-task. While unexpected, this finding might be caused by floor effects - distributions of the mental workload experiences show (see Figure 4A) that the overall experienced mental workload was relatively low, as was its variance.

No significant relationships were found between pre-task mental readiness and fatigue, and flow for either task. Interestingly, this finding contradicts previous observations that flow might be experienced less often during the day when recovery was poor [12]. For both tasks, a significant positive correlation between flow during the task and the level of mental readiness post-task could be seen. This indicates that being in flow might help to preserve mental readiness. As flow is described as a state of enjoyment, it is possible that being in flow has a positive impact on mood, motivation and concentration, which all positively influence mental readiness. The significant positive relationship between flow experience and post-task mental fatigue for the mental arithmetic task may be due to increased workload. Participants are more engrossed in the task and expend considerable cognitive resources when in flow, but also feel more fatigued after the task, pointing to a perpetuation of effort when mental readiness and flow coincide, even at the cost of increasing fatigue. Altogether, these results show that mental readiness and fatigue show different relationships with mental workload and flow, demonstrating divergent validity and suitability for different study purposes. Their utility for these different purposes warrants further research.

The described findings may also be evidence for the mental workload cycle introduced by Midha et al. [29]. Mental fatigue and

workload were studied among knowledge workers in a naturalistic setting, leading to the conclusion that people tend to prefer variation in workload levels. Coming of a low mental workload activity, participants wanted to engage in a high workload activity to spend their cognitive resources and feel productive. Thus, when participants report high mental readiness levels pre-task, they may feel that they are ready to be productive, expend cognitive resources, and fully immerse themselves in a task, explaining the positive effect on mental workload found in this study. This emphasizes that including mental readiness may lead to a better understanding for the dynamics of cognitive load during knowledge work.

Addressing the limitations of the presented study, the small sample size, skewed gender distribution, and duration have to be mentioned. An increased sample size with a balanced gender distribution would allow for a more precise and generalizable statistical examination of the covered concepts, while a more long-term, multi-week study would be able to more accurately capture workers cognitive load, and study long-term interactions of the concepts. Participants whose jobs cover tasks with a wide range of workload levels would need to be selected, to avoid the floor effects present in this study. Another key objective for future study should be to test whether mental readiness can be reliably detected through biosignals. For mental fatigue, workload, and flow, laboratory studies have already identified ways of detection through physiological signals, e.g., through EEG, during knowledge work [2, 22, 36]. Establishing the same for mental readiness would be a crucial step towards developing technology aimed at improving mental well-being at work, since self-assessments are often biased and inaccurate, e.g., people generally overrate their recovery from fatigue [23]. Biosignals on the other hand offer an objective way to detect cognitive phenomena and have already facilitated prototypical adaptive systems aimed at improving mental well-being at

work, like adjusting teaching speed based on mental workload [40], or break recommendation based on fatigue levels [24, 47].

The presented findings can be used to inform the development of future adaptive systems for fostering productivity and well-being in a work context. The relationships of mental readiness with flow and mental workload laid out in this paper may be used to enhance the predictive quality of such systems and allow for new approaches towards reaching desirable mental working conditions. In particular, an investigation into techniques aimed at fostering and maintaining mental readiness, like mental imagery or mindfulness breaks [27, 28], and whether high levels of mental readiness offer benefits in terms of increased productivity or flow during the workday, could be a promising next research step.

Acknowledgments

Funded by the German Research Foundation (Deutsche Forschungsgemeinschaft - DFG) – GRK2739/1 – Project Nr. 447089431 – Research Training Group: KD²School – Designing Adaptive Systems for Economic Decisions.

References

- [1] Daniel Afergan, Evan M. Peck, Erin T. Solovey, Andrew Jenkins, Samuel W. Hincks, Eli T. Brown, Remco Chang, and Robert J.K. Jacob. 2014. Dynamic difficulty using brain metrics of workload. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI '14). Association for Computing Machinery, New York, NY, USA, 3797–3806. doi:10.1145/2556288.2557230
- [2] Elizabeth M. Argyle, Adrian Marinescu, Max L. Wilson, Glyn Lawson, and Sarah Sharples. 2021. Physiological indicators of task demand, fatigue, and cognition in future digital manufacturing environments. *International Journal of Human-Computer Studies* 145, Article 102522 (2021).
- [3] Fabienne Aust, Theresa Beneke, Corinna Peifer, and Magdalena Wekenborg. 2022. The relationship between flow experience and burnout symptoms: A systematic review. *International Journal of Environmental Research and Public Health* 19, 7, Article 3865 (2022).
- [4] Jonathan Z. Bakdash and Laura R. Marusch. 2017. Repeated measures correlation. *Frontiers in Psychology* 8, Article 456 (2017).
- [5] Arnold B Bakker and Marianne van Woerkom. 2017. Flow at work: A self-determination perspective. *Occupational Health Science* 1 (2017), 47–65.
- [6] Karen Bartholomeyczik, Michael T. Knierim, and Christof Weinhardt. 2023. Fostering flow experiences at work: a framework and research agenda for developing flow interventions. *Frontiers of Psychology* 14, Article 1143654 (2023).
- [7] Karen Bartholomeyczik, Michael T. Knierim, Christof Weinhardt, and Gabriele Oettingen. 2024. Using mental contrasting to promote flow experiences at work: A just-in-time adaptive intervention. *Computers in Human Behavior Reports* 16, Article 100488 (2024).
- [8] Maarten A.S. Boksem, Theo F. Meijman, and Monique M. Lorist. 2005. Effects of mental fatigue on attention: An ERP study. *Cognitive Brain Research* 25, 1 (2005), 107–116.
- [9] Gianluca Borghini, Laura Astolfi, Giovanni Vecchiato, Donatella Mattia, and Fabio Babiloni. 2014. Measuring neurophysiological signals in aircraft pilots and car drivers for the assessment of mental workload, fatigue and drowsiness. *Neuroscience and Biobehavioral Reviews* 44 (2014), 58–75.
- [10] Adam Brown, Sarah D'Angelo, Ben Holtz, Ciera Jaspan, and Collin Green. 2023. Using Logs Data to Identify When Software Engineers Experience Flow or Focused Work. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [11] Mihaly Csikszentmihalyi and Judith LeFevre. 1989. Optimal Experience in Work and Leisure. *Journal of Personality and Social Psychology* 56, 4 (1989), 815–822.
- [12] Maike E. Debus, Sabine Sonnentag, Werner Deutsch, and Fridtjof W. Nussbeck. 2014. Making Flow Happen: The Effects of Being Recovered on Work-Related Flow Between and Within Days. *Journal of Applied Psychology* 99, 4 (2014), 713–722.
- [13] Evangelia Demerouti, Arnold B. Bakker, Friedhelm Nachreiner, and Margit Ebbinghaus. 2022. From mental strain to burnout. *European Journal of Work and Organizational Psychology* 11, 4 (2022), 423–441.
- [14] Stefan Engeser and Falko Rheinberg. 2008. Flow, performance and moderators of challenge-skill balance. *Motivation and Emotion* 32, 3 (2008), 158–172.
- [15] Jialin Fan and Andrew P. Smith. 2019. Mental workload and other causes of different types of fatigue in rail staff. In *Communications in Computer and Information Science*. Springer International Publishing, 147–159.
- [16] Shree Frazier, Brandon J. Pitts, and Sara McComb. 2022. Measuring cognitive workload in automated knowledge work environments: a systematic literature review. *Cognition, Technology and Work* 24, 4 (2022), 557–587.
- [17] Clive J. Fullagar and E. Kevin Kelloway. 2009. Flow at work: An experience sampling approach. *Journal of Occupational and Organizational Psychology* 82, 3 (2009), 595–615.
- [18] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. *Advances in Psychology* 52, 1 (1988), 139–183.
- [19] Mega B. Herlambang, Niels A. Taatgen, and Fokke Cnossen. 2019. The role of motivation as a factor in mental fatigue. *Human Factors* 61, 7 (2019), 1171–1185.
- [20] Kenji Katahira, Yoichi Yamazaki, Chiaki Yamaoka, Hiroaki Ozaki, Sayaka Nakagawa, and Noriko Nagata. 2018. EEG correlates of the flow state: A combination of increased frontal theta and moderate frontocentral alpha rhythm in the mental arithmetic task. *Frontiers of Psychology* 9 (2018).
- [21] Johannes Keller, Herbert Bless, Frederik Blomann, and Dieter Kleinbühl. 2011. Physiological aspects of flow experiences: Skills-demand-compatibility effects on heart rate variability and salivary cortisol. *Journal of Experimental Social Psychology* 47, 4 (2011), 849–852.
- [22] Michael Thomas Knierim, Daniel Puhl, Gabriel Ivucic, and Tobias Röddiger. 2023. OpenBCI + 3D-printed headphones = open ExG headphones – an open-source research platform for biopotential wearable applications. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg Germany). ACM, New York, NY, USA, 1–7.
- [23] Justine R Magnuson, Sam M Doesburg, and Chris J McNeil. 2021. Development and recovery time of mental fatigue and its impact on motor function. *Biological Psychology* 161, Article 108076 (2021).
- [24] Moritz Maleck and Tom Gross. 2024. CoLoTiMa: A cognitive-load based time management tool. In *Proceedings of Mensch und Computer 2024* (Karlsruhe Germany). ACM, New York, NY, USA, 690–694.
- [25] Nicki Marquardt, Carolin Hannig, and Sandra Hannig. 2018. The impact of mental readiness on driving performance and traffic safety. *International Journal of Safety and Security Engineering* 8, 1 (2018), 10–19.
- [26] Judy M. McDonald. 2021. Mental readiness: focusing on the cutting edge. *Archives of Pediatric Neurosurgery* 3, 2, Article e952021 (2021).
- [27] Judy M. McDonald and Terry Orlick. 1995. Mental Readiness in Surgeons and Its Links to Performance Excellence in Surgery. *Journal of Pediatric Orthopaedics* 15 (1995), 691–697.
- [28] Judy M. McDonald and Corrado Paganelli. 2021. Exploration of Mental Readiness for Enhancing Dentistry in an Inter-Professional Climate. *International Journal of Environmental Research and Public Health* 18, Article 7038 (2021).
- [29] Serena Midha, Max L. Wilson, and Sarah Sharples. 2022. Lived Experiences of Mental Workload in Everyday Life. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 282, 16 pages. doi:10.1145/3491102.3517690
- [30] Alyssa S.C. Ng, Stijn A.A. Massar, Bei Bei, and Michael W.L. Chee. 2023. Assessing 'readiness' by tracking fluctuations in daily sleep duration and their effects on daily mood, motivation, and sleepiness. *Sleep Medicine* 112 (2023), 30–38.
- [31] Terry Orlick and John Partington. 1988. Mental Links to Excellence. *The Sport Psychologist* 2 (1988), 105–130.
- [32] O. Pastushenko, Wilk Oliveira, Seiji Isotani, and Tomás Hruska. 2020. A Methodology for Multimodal Learning Analytics and Flow Experience Identification within Gamified Assignments. *CHI Extended Abstracts* (2020). doi:10.1145/3334480.3383060
- [33] Corinna Peifer, Gina Wolters, László Harmat, Jean Heutte, Jasmine Tan, Teresa Freire, Dionisia Tavares, Carla Fonte, Frans Orsted Andersen, Jef van den Hout, Milija Šimleša, Linda Pola, Lucia Ceja, and Stefano Triberti. 2022. A scoping review of flow research. *Frontiers of Psychology* 13, Article 815665 (2022).
- [34] André Pimenta, Davide Carneiro, José Neves, and Paulo Novais. 2016. A neural network to classify fatigue from human-computer interaction. *Neurocomputing* 172 (2016), 413–426.
- [35] René Riedl, Kyriaki Kostoglou, Selina C. Wriessnegger, and Gernot Müller-Putz. 2023. Videoconference fatigue from a neurophysiological perspective: experimental evidence based on electroencephalography (EEG) and electrocardiography (ECG). *Scientific Reports* 13, 1, Article 18371 (2023).
- [36] Raphael Rissler, Mario Nadj, Maximilian X. Li, Michael Thomas Knierim, and Alexander Maedche. 2018. Got flow? Using Machine Learning on Physiological Data to Classify Flow. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC Canada). ACM, New York, NY, USA.
- [37] Raphael Rissler, Mario Nadj, Maximilian X. Li, Nico Loewe, Michael T. Knierim, and Alexander Maedche. 2023. To Be or Not to Be in Flow at Work: Physiological Classification of Flow Using Machine Learning. *IEEE Transactions on Affective Computing* 14, 1 (2023), 463–474.
- [38] Philip Schmidt, Attila Reiss, Robert Dürichen, and Kristof Van Laerhoven. 2019. Wearable-based affect recognition-A review. *Sensors* 19, 19, Article 4079 (2019).

- [39] E M Smets, B Garssen, B Bonke, and J C De Haes. 1995. The Multidimensional Fatigue Inventory (MFI) psychometric qualities of an instrument to assess fatigue. *Journal of Psychosomatic Research* 39, 3 (1995), 315–325.
- [40] Chiew Seng Sean Tan, Johannes Schöning, Kris Luyten, and Karin Coninx. 2014. Investigating the effects of using biofeedback as visual stress indicator during video-mediated collaboration. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto Ontario Canada). ACM, New York, NY, USA.
- [41] Isaac Teoh Yi Zhe and Pantea Keikhosrokiani. 2021. Knowledge workers mental workload prediction using optimised ELANFIS. *Applied Intelligence* 51, 4 (2021), 2406–2430.
- [42] Mindaugas Vasiljevas, Robertas Damaševičius, and Rytis Maskeliūnas. 2023. A human-adaptive model for user performance and fatigue evaluation during gaze-tracking tasks. *Electronics* 12, 5, Article 1130 (2023).
- [43] Vilma Vuori, Nina Helander, and Jussi Okkonen. 2019. Digitalization in knowledge work: the dream of enhanced performance. *Cognition, Technology and Work* 21, 2 (2019), 237–252.
- [44] Novia Wong, Victoria Jackson, André Van Der Hoek, Iftekhar Ahmed, Stephen M. Schueller, and Madhu Reddy. 2023. Mental Wellbeing at Work: Perspectives of Software Engineers. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 858, 15 pages. doi:10.1145/3544548.3581528
- [45] Mark S Young, Karel A Brookhuis, Christopher D Wickens, and Peter A Hancock. 2015. State of science: mental workload in ergonomics. *Ergonomics* 58, 1 (2015), 1–17.
- [46] Yinhu Yu, Anastasios Bezerianos, Andrzej Cichocki, and Junhua Li. 2023. Latent Space Coding Capsule Network for Mental Workload Classification. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 31 (2023), 3417–3427.
- [47] Ramtin Zargari Marandi, Pascal Madeleine, Øyvind Omland, Nicolas Vuillerme, and Afshin Samani. 2019. An oculometrics-based biofeedback system to impede fatigue development during computer work: A proof-of-concept study. *PLoS One* 14, 5, Article e0213704 (2019).
- [48] Qiushi Zhou, Joshua Newn, Namrata Srivastava, Tilman Dingler, Jorge Goncalves, and Eduardo Velloso. 2019. Cognitive aid: task assistance based on mental workload estimation. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–6.
- [49] Manuela Züger, Christopher Corley, André N Meyer, Boyang Li, Thomas Fritz, David Shepherd, Vinay Augustine, Patrick Francis, Nicholas Kraft, and Will Snipes. 2017. Reducing interruptions at work: A large-scale field study of flowlight. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 61–72.